

Homework

Basics in linear algebra and numerical methods, 2025/26. semester I.

General instructions: Each week, you have to submit the solution of three arbitrary exercises. This should be done as a Python code. Comments should be written in English.

Scores: 3×3 per week; altogether: 54 in the whole semester.

Week 2

- (a) Read the RGB code of a picture and apply left-right mirroring.
- (b) Take the numbers $1, 2, \dots, 12$. Partition them into 2 groups such that each group has six elements. Print all different partitions in a matrix, where one row shows one partition.
- (c) Take a random array of size $6 \times 6 \times 6$. Sort their layers such that the minimums are in increasing order: minimum of new layer $0 \leq$ minimum of new layer $1 \leq \dots \leq$ minimum of new layer 5.
- (d) Take a stochastic matrix of size 20×10 . Print the row which has minimal average distance from the other row vectors.
- (e) Given 2 vectors $u = (3, 0, 1)$ and $v = (1, 1, 1)$, find a matrix that projects any vector in $\mathbb{R}^3$ to the subspace generated by these 2 vectors. How would the projection of a sphere of radius 1, centered at 0, would look like?

Hint: Gram-Schmidt!